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CONTAINER ASSEMBLY

Abstract

A container assembly has a thermoformed plastic container with a base and a container wall terminating at a container flange, and a flexible container lid is attached to an upper surface of the container flange by a first heat sealing material. The assembly also includes a support with a support wall surrounding the container wall and terminating at a support flange, wherein the support flange is configured and arranged to abut against a lower surface of the container flange, and is attached thereto by a second heat sealing material.

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Background/Summary

FIELD

[0001] The present teachings relate to a container assembly, and to a method of forming a container assembly.

BACKGROUND

[0002] Thermoformed plastic containers are often used for packaging for a range of different product types and in a range of different sizes and shape. These thermoformed plastic containers are suitable for the storage and transportation of liquid or semi-solid foodstuffs such as, for example, soups, sauces, yoghurt and other dairy products, or solid foodstuffs such as, for example, nuts, dried fruit pieces and the like. These containers are also used to store and transport meat and fish, where modified atmosphere packaging (MAP) is often used to extend shelf life.

[0003] Despite rising awareness among consumers of global environmental factors such as global warming and rising marine pollution, the trend for consuming food and drinks from plastic containers is growing rapidly. Thus, the amount of plastics material, and so the raw materials such as oil, used to meet this demand is also increasing. This vast usage of raw materials results in a large carbon footprint for the plastics industry, thus contributing to global warming.

[0004] The present teachings seek to overcome or at least mitigate one or more problems associated with the prior art.

SUMMARY

[0005] According to a first aspect, there is provided a container assembly comprising: a thermoformed plastic container comprising a base, and a container wall extending from and surrounding the base and terminating at a container flange extending around the container wall to define a mouth of the container; a support comprising a support wall arranged to surround the container wall, said support wall terminating at a support flange; and a flexible container lid attached to an upper surface of the container flange by a first heat sealing material, wherein the support flange is configured and arranged to abut against a lower surface of the container flange, and wherein the support flange is attached to the lower surface of the container flange by a second heat sealing material.

[0006] The support provided by the support to the thermoformed plastic container enables a reduced amount of plastic material to be used to form the container. Attachment of the flexible container lid and the support to opposing upper and lower surfaces of the container flange enables the flexible container lid and the support to be secured to the container flange in a single step by applying heat and pressure to the top and bottom of the container flange. This facilitates the assembly process for the container assembly.

[0007] In some embodiments, heat may be applied to the upper surface of the container flange to heat the upper and lower surfaces of the container flange. Put another way, heat may be applied directly to the upper surface of the container flange and indirectly to the lower surface of the container flange through said container flange. In such embodiments, the melting temperature of the second heat seal material may be lower than that of the first heat sealing material.

[0008] In other embodiments, heat may be applied to the lower surface of the container flange to heat the upper and lower surfaces of the container flange. Put another way, heat may be applied directly to the lower surface of the container flange and indirectly to the upper surface of the container flange through said container flange. In such embodiments, the melting temperature of the first heat seal material may be lower than that of the second heat sealing material.

[0009] In other embodiments, heat may be applied directly to the upper and lower surfaces of the container flange.

[0010] The support wall may abut against the container wall to provide support thereto. The

tapered container wall may be seated on the tapered support wall. The container flange may be seated on the support flange.

[0011] The container flange may comprise a downwardly projecting lip extending from a peripheral edge of the container flange in a direction towards the base, and the lip may oppose the container wall to define a channel therebetween receiving the support flange therein. This arrangement has been found to facilitate assembly of the support onto the container as the channel acts as a locator for the support flange.

[0012] The support flange may be mechanically secured between the lip and the container wall. This arrangement facilitates assembly of the support and container prior to heat sealing the support and container by providing a mechanical attachment therebetween.

[0013] In some embodiments, an adhesive may be applied to the lower surface of the container flange and/or the upper surface of the support flange. The adhesive may be applied as one or more spots of adhesive on the container flange and/or support flange. The use of adhesive may help to hold the base of the container to the base of the support prior to heat sealing the support and container.

[0014] The support and the container may be configured and arranged to mechanically fix the support to the container. This arrangement has been found to facilitate assembly of the support and container prior to the attachment via the heat sealing materials.

[0015] The support may comprise a tab projecting from a distal edge of the support flange so as to extend beyond a distal end of the lip. This facilitates removal of the support from the container for separate recycling, after the container assembly has been used.

[0016] The container wall may comprise a plurality of container wall panels defining corner regions at an intersection between adjacent container wall panels and the base, and wherein each corner region is tapered or chamfered. This reduces the amount of plastic that is required to form the plastic container as it reduces the distance from a centre point of the container to the corners.

[0017] The support wall may comprise a plurality of support wall panels, and wherein the support flange comprises a support flange part extending from a distal edge of each support wall panel. The provision of the support flange being formed by multiple separate flange parts has been found to facilitate assembly of the support and container.

[0018] The support flange parts may be configured and arranged such that they form a continuous support flange around the support wall. Alternatively, the support flange parts may be configured and arranged such that they do not form a continuous support flange around the support wall.

[0019] Each support flange part may be tapered or chamfered at opposing ends thereof. This has been found to facilitate assembly of the support and container. In some embodiments, each support flange part may be tapered or chamfered at an angle in the range 40° - 50° at opposing ends thereof.

[0020] The container wall may be substantially linear in a direction between the base and the container flange and tapers in a direction towards the base. In some embodiments, the container wall may taper at an angle in the range 5° to 19° , in the range 10° to 18° , in the range 12° to 16° , for example approximately 14° . In some embodiments, the container wall may taper at an angle in the range 4° to 18° , in the range 8° to 16° , in the range 10° to 14° , for example approximately 12° . Providing angles in these ranges has been found to improve stacking of the containers side by side.

[0021] The support wall may conform to the shape of the container wall. This helps to improve the support provided by the support to the container.

[0022] The flexible lid may be formed from a plastic film, a metal foil, a paper material, and/or a card material. In some embodiments the flexible lid may be coated with the first heat sealing material.

[0023] The first heat sealing material may be selected from: a non-plastic heat sealable material such as polylactic acid; an aqueous heat-sealing material; and/or a polymeric film such as polyethylene, polypropylene, polyethylene terephthalate.

[0024] In some embodiments, the second heat sealing material may be selected from: a non-plastic

heat sealable material such as polylactic acid; an aqueous heat-sealing material; and/or a polymeric film such as polyethylene, polypropylene, and/or polyethylene terephthalate.

[0025] The support may be formed from a fibrous material such as paper or card.

[0026] In some embodiments, the support may be formed from a folded and glued blank, formed for example from paper or card. In alternatively embodiments, the support may be a pre-formed pulp board container (like high quality egg boxes).

[0027] The support flange may be coated with the second heat sealing material.

[0028] The support wall may comprise a plurality of support wall panels defining corner regions at an intersection between adjacent container wall panels. The support flange may extend continuously around support wall or may not extend continuously around the support wall. In either case, the support flange may extend around, or into, the corners of the support wall. Put another way, the corner regions of the support wall may terminate at the support flange.

[0029] The container wall may comprise eight container wall panels. The support wall may also comprise eight support wall panels.

[0030] According to a second aspect, there is provided a method of forming a container assembly, the method comprising the steps of: thermoforming a plastic container comprising a base, and a container wall extending from and surrounding the base and terminating at a container flange extending around the container wall to define a mouth of the container; providing a support comprising a support wall terminating at a support flange; positioning the container within the support such that the support wall surrounds the container wall and the support flange abuts against a lower surface of the container flange; positioning a flexible container lid over an upper surface of the container flange; and applying heat and pressure to the upper and lower surfaces of the container flange to attach the flexible container lid to the upper surface of the container flange and the support flange to the lower surface of the container flange via first and second heat sealing materials, respectively.

[0031] The step of applying heat and pressure to the upper and lower surfaces of the container flange may comprise simultaneously applying heat and pressure to the upper and lower surfaces of the container flange.

[0032] In some embodiments, heat may be applied to the upper surface of the container flange to heat the upper and lower surfaces of the container flange. Put another way, heat may be applied directly to the upper surface of the container flange and indirectly to the lower surface of the container flange through said container flange. In such embodiments, the melting temperature of the second heat seal material may be lower than that of the first heat sealing material.

[0033] In other embodiments, heat may be applied to the lower surface of the container flange to heat the upper and lower surfaces of the container flange. Put another way, heat may be applied directly to the lower surface of the container flange and indirectly to the upper surface of the container flange through said container flange. In such embodiments, the melting temperature of the first heat seal material may be lower than that of the second heat sealing material.

[0034] In other embodiments, heat may be applied directly to the upper and lower surfaces of the container flange.

[0035] The method may comprise the step of forming the flexible container lid from the first heat seal material.

[0036] The method may comprise the step of coating the flexible container lid and/or the upper surface of the container flange with the first heat seal material.

[0037] The method may comprise the step of coating the lower surface of the container flange and/or the support flange with the second heat seal material.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0038] Embodiments will now be described with reference to the accompanying drawings, in which:

[0039] FIG. 1 is an exploded isometric view of a container assembly according to an embodiment;

[0040] FIG. 2 is a cross-sectional side view of the container assembly of FIG. 1;

[0041] FIG. 3 is an enlarged cross-sectional side view of the circled region of FIG. 2;

[0042] FIG. 4 is a plan view of the support of the container assembly of FIG. 1, prior to assembly;

[0043] FIG. 5 is an exploded isometric view of a container assembly according to an embodiment;

[0044] FIG. 6 is a plan view of the support of the container assembly of FIG. 5, prior to assembly; and

[0045] FIG. 7 is a plan view of the container of the container assembly of FIG. 5.

DETAILED DESCRIPTION OF EMBODIMENT(S)

[0046] Referring to FIGS. 1 to 3, there is shown a container assembly generally designated **10**. The container assembly **10** includes a plastic container **12** and a support **14**. The container **12** may be a thermoformed plastic container. The container **12** is positioned in and is supported by the support **14**. The support provided to the thermoformed plastic container **12** enables a reduced amount of plastic material to be used to form the container **12**.

[0047] The container **12** has a base **16** and a container wall **18** surrounding the base **16**. The container wall **18** extends from the base **16**. The container wall **18** extends to and terminates at a container flange **20** extending around the periphery of the container **12**. Put another way, the container flange **20** extends around the periphery of the container wall **18**. The container flange **20** extends around an entirety of the periphery of the container **12** (i.e. around an entirety of the container wall **18**). The base **16** and container wall **18** co-operate to define an interior space of the container **12**. The container flange **20** defines a mouth **24** of the container **12**.

[0048] The container **12** may be manufactured from sheet polyethylene terephthalate (PET). The container **12** may alternatively be manufactured from sheet polypropylene (PP), from crystallised polyester (CPET), or from amorphous polyethylene terephthalate (APET). It will be understood that the term “thermoforming” refers to the forming of the container by the thermal softening of a sheet of plastic material, which is then conformed to the shape of a mould. In alternative arrangements, any suitable plastics material may be used. It will be appreciated that the sheet material may have a thickness of less than 300 microns, for example in the range 150 to 250 microns, for example approximately 200 microns. Providing a sheet of these thicknesses have been found to ensure that the thickness of the resulting container is sufficient (e.g. at least 100 microns) over the entire container.

[0049] As an alternative, the container **12** may be formed from sheet RPET that can contain up to 100% recycled post-consumer waste. In yet a further embodiment, the container **10** may be formed from laminated PET/RPET having an A/B/A structure with 100% RPET in the central layer and two outer layers of PET. Any thickness combination is possible. For a sheet thickness of 1000 micron, the layers may be as follows—with 100 PET/800 RPET/100 PET.

[0050] The support **14** is fitted to the container **12** to form the container assembly **10**. The support **14** includes a support wall **26** that surrounds the container wall **18**. Put another way, the support wall **26** defines an opening into which the container **12** is positioned. The support wall **26** terminates at a support flange **28**. The support flange **28** defines a mouth **30** or upper opening of the support **14**. In the illustrated arrangement, the support **14** includes a base **32** such that the support **14** may be considered to be a tray. In alternative arrangements, the base **32** may be omitted and the support **14** may be considered to be a sleeve. The support **14** may be formed from a fibrous material such as paper or card. The support **14** (i.e. the tray or sleeve) may be provided as a decorative card. The card which forms the support **14** may have a thickness of less than 650 microns, for example less than 500 microns, for example, less than 400 microns, for example in the

range 250 to 400 microns, for example in the range 300 to 350 micron, for example in the range 330 to 340 microns, for example approximately 335 microns.

[0051] The container flange **20** includes a substantially flat or planar upper surface **34** that surrounds the mouth **24** of the container **12**. The upper surface **34** defines an attachment location for a flexible container lid **36**. The plane of the upper surface **34** is substantially parallel to the plane of the base **16** of the container **12**. The container flange **20** includes a peripheral edge **38**. The peripheral edge **38** is defined by a lip **40** extending in a direction towards the base **16**. The peripheral edge **38** of the container flange **20** may be used to retain a rigid replaceable plastic lid (not shown) that is configured to fit over the container flange **20**.

[0052] The flexible container lid **36** may be formed from a plastic film, a metal foil, paper or card. The flexible container lid **36** is attached to the upper surface **34** of the container flange **20** by a first heat sealing material (not shown). The first heat sealing material may be selected from a polymeric film such as polyethylene, polypropylene, polyethylene terephthalate, and/or a metal foil such as aluminium foil. It will be appreciated that the flexible container lid may be formed from the first heat sealing material, or may be coated with the first heat sealing material. Alternatively, it will be understood that the container flange may be coated with the first heat sealing material. The flexible container lid **36** is attached to the upper surface **34** by the application of heat and pressure.

[0053] The container flange **20** includes a substantially flat or planar lower surface **42**. The lower surface **42** defines an underside of the container flange **20**. The support flange **28** is configured and arranged to abut against the lower surface **42** of the container flange **20**, as is shown in FIG. 3. Put another way, the container flange is seated on the support flange. The support flange **28** is attached to the lower surface **42** of the container flange **20** by a second heat sealing material (not shown). The second heat sealing material may be a non-plastic heat sealable material, such as polylactic acid. In some arrangements, the second heat sealing material may be an aqueous heat sealing material, for example aqueous dispersions of low-density polyethylene and/or ethylene vinyl acetate. In some arrangements, the second heat sealing material may be polymeric film such polyethylene, polypropylene, polyethylene terephthalate. The second heat sealing material may be the same or different to the first heat sealing material. It will be appreciated that the support flange **28** may be coated with the second heat sealing material, or the lower surface **42** of the container flange **20** may be coated with the second heat sealing material. The support **14**, i.e. the support flange **28**, is attached to the lower surface **42** by the application of heat and pressure.

[0054] The flexible container lid **36** and the support flange **28** are attached to opposing upper and lower surfaces **34**, **42** of the container flange **20** via heat sealing materials. This arrangement enables the flexible container lid **36** and the support flange **28** to be secured to the container flange **20** in a single step by applying heat and pressure to the upper and lower surfaces **34**, **42** of the container flange **20**.

[0055] In some embodiments, heat may be applied to the upper surface of the container flange to heat the upper and lower surfaces of the container flange. Put another way, heat may be applied directly to the upper surface of the container flange and indirectly to the lower surface of the container flange through said container flange. In such embodiments, the melting temperature of the second heat seal material may be lower than that of the first heat sealing material. In other embodiments, heat may be applied to the lower surface of the container flange to heat the upper and lower surfaces of the container flange. Put another way, heat may be applied directly to the lower surface of the container flange and indirectly to the upper surface of the container flange through said container flange. In such embodiments, the melting temperature of the first heat seal material may be lower than that of the second heat sealing material. In other embodiments, heat may be applied directly to the upper and lower surfaces of the container flange.

[0056] As discussed above, the container flange **20** includes a lip **40** extending from a peripheral edge **38** of the container flange **20** in a direction towards the base **16**. The support **14** a tab **46** projecting from a distal edge **48** of the support flange **28**. The tab **46** is provided to enable a user to

separate the support **14** from the container **12** after the container assembly **10** has been used. The tab **46** is arranged so as to extend beyond a distal end **48** of the lip **40**.

[0057] The lip **40** is arranged to oppose the container wall **18** to define a channel **44** therebetween. The channel **44** received the support flange **28** therein. The support flange **28** may be mechanically secured to the container **12**. Put another way, the support flange **28** may be mechanically secured between the lip **40** and the container wall **18**. The support flange **28** may be dimensioned such that it abuts against the lip **40** and the container wall **18** to mechanically secure the support **14** to the container **12**. In some embodiments, an adhesive may be applied, e.g. as one or spots, to the lower surface of the container flange and/or the upper surface of the support flange to hold the base of the container to the base of the support prior to heat sealing the support and container.

[0058] In the illustrated arrangement, the container **12** and the support **14** are generally rectangular in plan-view. The container **12** and support **14** each include four container wall panels or side walls. The wall panels are substantially planar and are arranged to form the container wall **18** and support wall **26**, respectively. In alternative arrangements, it will be appreciated that the container **12** and support **14** may be any suitable shape in plan-view such as circular, triangular, rectangular, pentagonal, hexagonal, octagonal or any other suitable shape. It will be appreciated that the container **12** may have other shapes and volumes to suit the application.

[0059] The container wall **18** is substantially linear in a direction between the base **16** and the container flange **20**. The container wall **18** tapers in a direction towards the base **16**. Put another way, the container wall **18** defines a width that reduces from a maximum width at the container flange **20** to a minimum width at the base **16**. This tapered wall configuration enables empty containers **12** to be nested within one another for storage and transportation. The container wall **18** tapers at an angle in the range 5° to 19° , in the range 10° to 18° , in the range 12° to 16° , for example approximately 14° . Alternatively, the container wall **18** may taper at an angle in the range 4° to 18° , in the range 8° to 16° , in the range 10° to 14° , for example approximately 12° .

[0060] The shape of the support wall **26** conforms to the shape of the container wall **18**. Put another way, the support wall **26** is substantially linear and tapers in a direction away from the support flange **28**. In this way, the support wall **26** abuts against the container wall **18** to provide support thereto. Put another way, the tapered container wall **18** is seated on the tapered support wall **26**. It will be appreciated that the features of the container wall **18** described above are also applicable to the support wall **26**. Providing a support wall **26** that conforms to the shape of the container wall **18** has been found to improve the support provided to the container **12**.

[0061] The container wall **18** is formed from four container side walls or container wall panels extending around the base **16**. In this way, the container **12** defines corner regions **50** at the intersection between adjacent container side walls and the base **16**. Each corner region **50** is tapered or chamfered. This arrangement reduces the distance from a centre point of the container **12** to the corner regions **50** of the container **12**, which reduces the thickness of plastic material needed to thermoform the container **12**.

[0062] Referring now to FIG. **4**, the support **14** is provided in the form of a foldable blank that is to be assembled prior to fitting to the container **12**. In this arrangement, the support wall **26** is formed from four support wall panels or support side wall panels **54**. It will be appreciated that in alternative arrangements any suitable number of wall panels **54** may be provided. Each wall panel **54** is connected to and extends from the base **32**, but in arrangements where the base **32** is omitted it will be understood that one or more of the wall panels **54** may be foldably connected.

[0063] The support may be provided with one or more attachment flaps **56** to secure adjacent support wall panels together to form the support wall **26**. Attachment may be achieved via an adhesive or any other suitable means. In the illustrated embodiment, the end wall panels are each provided with two attachment flaps to secure the two adjacent side wall panels thereto. It will be understood, however, that any suitable arrangement of the attachment flaps may be used to secure

the support wall panels together to form the support wall **26**. In the present arrangement, one or more of the wall panels **54** are provided with an attachment flap **56** extending from a lateral edge of the wall panel **54** and arranged to contact an adjacent wall panel **54** during assembly of the support **14**. It will be understood that the attachment flap(s) **56** will be attached to the adjacent wall panel **54** via the adhesive, which may be applied or coated on the attachment flap **56** and/or the adjacent wall panel **54**.

[0064] The support flange **28** is provided as a support flange part **52** extending from a distal edge of each support side wall panel **54**. The provision of the support flange **28** being formed from multiple separate flange parts has been found to facilitate assembly of the support **14** and container **12**. In the illustrated arrangement, the support flange parts **52** are configured and arranged such that they do not form a continuous flange around the support wall **26**. In the arrangement shown, each support flange part **52** is tapered at opposing ends thereof. The taper at each end of the support flange parts **52** may be at an angle in the range 30°-60°, for example in the range 40°-50°, for example approximately 45°. In some alternative arrangements, each flange support part **52** may not be tapered and may extend substantially perpendicularly from the distal edge of the respective side wall panel **54**. In further alternative arrangements, the support flange parts **52** may be configured and arranged to form a continuous flange around the support wall **26**.

[0065] Referring now to FIGS. 5 to 7, a second embodiment of the container assembly **10** is shown. Like the container assembly **10** of the first embodiment, the container assembly **10** of the second embodiment comprises a container **12** and a support **14**. The container **12** may be a thermoformed plastic container, as has been described with reference to FIGS. 1 to 3. The container **12** is positioned in and supported by the support **14**. The support **14** may be formed from a fibrous material such as paper or card. The second embodiment has many elements in common with the container assembly of FIGS. 1 to 4, and hence, in the following, only differences will be described in detail.

[0066] The container **12** has a base **16** and a container wall **18** surrounding the base **16**. The container wall **18** extends from the base **16**. The container wall **18** extends to and terminates at a container flange **20** extending around the periphery of the container **12**. Put another way, the container flange **20** extends around the periphery of the container wall **18**. The container flange **20** extends around an entirety of the periphery of the container **12** (i.e. around an entirety of the container wall **18**). The base **16** and container wall **18** co-operate to define an interior space of the container **12**. The container flange **20** defines a mouth **24** of the container **12**.

[0067] The container flange **20** includes a lip **40** extending from a peripheral edge **38** of the container flange **20** in a direction towards the base **16**. The support **14** has a tab **46** projecting from a distal edge **48** of the support flange **28**. The tab **46** is provided to enable a user to separate the support **14** from the container **12** after the container assembly **10** has been used. The tab **46** is arranged so as to extend beyond a distal end **48** of the lip **40**.

[0068] The lip **40** is arranged to oppose the container wall **18** to define a channel **44** therebetween. The channel **44** receives the support flange **28** therein. The support flange **28** may be mechanically secured to the container **12**. Put another way, the support flange **28** may be mechanically secured between the lip **40** and the container wall **18**. The support flange **28** may be dimensioned such that it abuts against the lip **40** and the container wall **18** to mechanically secure the support **14** to the container **12**. In some embodiments, an adhesive may be applied, e.g. as one or spots, to the lower surface of the container flange and/or the upper surface of the support flange to hold the base of the container to the base of the support prior to heat sealing the support and container.

[0069] The container **12** of the second embodiment includes eight wall panels. These include a first pair of equal and opposite lengthwise wall panels which extend parallel to the length of the container **12**. Put another way, the container includes a first pair of side wall panels. The container **12** includes a second pair of equal and opposite width-wise wall panels which extend parallel to the

width of the container **12**. Put another way, the container **12** include a second pair of end wall panels. The container **12** includes four corner wall panels which extend between and connect the lengthwise and widthwise wall panels to form a continuous container wall. Put another way, each corner container wall panel extending between a side wall panel and an adjacent end wall panel. The side wall panels define a substantially octagonal container. The side wall panels are arranged perpendicular to the end wall panels, and each corner wall panel extends at an angle between the adjacent end wall and side wall of the container. Each of the corner walls panels spans a smaller distance than either the lengthwise or widthwise wall panels, such that the container **12** substantially describes a rectangle with chamfered corners in plan-view. Similar to the container of FIGS. **1** to **3**, the container wall **18** extends to and terminates at a container flange **20** which extends around the periphery of the container **12** and which defines a mouth **24** of the container **12**. The container flange defines a perimeter that is substantially rectangular, for example the perimeter defines a rectangle with curved corners.

[0070] The support **14** of the second embodiment includes eight wall panels. These include a first pair of equal and opposite lengthwise wall panels which extend parallel to the length of the support **14**. Put another way, the support includes a first pair of side support wall panels. The support **14** includes a second pair of equal and opposite width-wise wall panels which extend parallel to the width of the support **14**. Put another way, the support **14** includes a second pair of end support wall panels. The support **14** includes four corner wall panels which extend between and connect the lengthwise and widthwise wall panels to form a continuous support wall. Put another way, the support includes four corner support wall panels each corner support wall panels extending a support side wall panel and an adjacent support end wall panel. The support side wall define a substantially octagonal support **14**. The side wall panels are arranged perpendicular to the end wall panels, and each corner wall panel extends at an angle between the adjacent end wall and side wall of the container. Each of the corner walls panels spans a smaller distance than either the lengthwise or widthwise wall panels, such that the support **14** substantially describes a rectangle with chamfered corners in plan-view. Similar to the support **14** of FIGS. **1** to **4**, the support wall **26** terminates at a support flange **28** which defines a mouth **30** or upper opening of the support **14**. The support flange defines a perimeter that is substantially rectangular, for example the perimeter defines a rectangle with curved corners.

[0071] As described above in connection with the container assembly of FIGS. **1** to **4**, the support flange **28** is attached to the lower surface **42** of the container flange **20** by a second heat sealing material (not shown). The second heat sealing material may be a non-plastic heat sealable material, such as polylactic acid. In some arrangements, the second heat sealing material may be an aqueous heat-sealing material, for example aqueous dispersions of low-density polyethylene and/or ethylene vinyl acetate. In some arrangements, the second heat sealing material may be polymeric film such polyethylene, polypropylene, polyethylene terephthalate. The second heat sealing material may be the same or different to the first heat sealing material. It will be appreciated that the support flange **28** may be coated with the second heat sealing material, or the lower surface **42** of the container flange **20** may be coated with the second heat sealing material. The support **14**, i.e. the support flange **28**, is attached to the lower surface **42** by the application of heat and pressure.

[0072] Referring now to FIG. **6**, the foldable blank used to form the support **14** of the second embodiment includes eight support wall panels or support side wall panels **54**. The support side wall panels **54** include four corner wall panels **54A** which define the chamfered corners of the rectangle described by the support **14** in plan-view when the foldable blank is assembled into the support **14**. In the arrangement shown, the attachment flaps **56** which join the support wall panels **54** to form a continuous support wall are provided on the lateral edges of the corner wall panels **54A**.

[0073] The support may be provided with one or more attachment flaps **56** to secure adjacent support wall panels together to form the support wall **26**. Attachment may be achieved via an

adhesive or any other suitable means. In the illustrated embodiment, the corner wall panels **54A** are each provided with two attachment flaps to secure the adjacent side and end wall panels thereto. It will be understood, however, that any suitable arrangement of the attachment flaps may be used to secure the support wall panels together to form the support wall **26**. In the present arrangement, one or more of the wall panels **54** are provided with an attachment flap **56** extending from a lateral edge of the wall panel **54** and arranged to contact an adjacent wall panel **54** during assembly of the support **14**. It will be understood that the attachment flap(s) **56** will be attached to the adjacent wall panel **54** via the adhesive, which may be applied or coated on the attachment flap **56** and/or the adjacent wall panel **54**.

[0074] As described above in connection with the support **14** of FIG. **4**, the support flange **28** is provided as a series of first support flange parts **52** that form the support flange when the support **14** is assembled. In the embodiment shown, a first support flange part **52** extends from a distal edge of each of the end and side support wall panels **54**.

[0075] In some embodiments, a second support flange part **52A** extends from a distal edge of each of the corner wall panels **54A**. The second support flange parts **52A** provide support to the corner regions of the container flange **20** when the support flange **58** is attached to the container flange **20** under pressure. The second support flange parts **52** help to provide a uniform seal around the container flange and may increase the strength of the assembled sealed container assembly **10**. The second support flange parts **52A** may define a curved outer. The first support flange parts may define a substantially linear outer edge. The term “outer” here refers to the edge of the support flange parts **52, 52A** furthest away from the mouth **30**. In the arrangement shown, the support flange parts **52** are configured and arranged to form a continuous flange around the support wall **26**. In alternative embodiments, the second support flange parts may be omitted. In such embodiments, the first support flange parts **52** may project beyond the respective end or side support wall panels **54** so as to define a continuous flange around the support wall **26**. This arrangement reduces the number of breaks in the container flange, which has been found to increase the strength of the container flange, and of the assembled container assembly **10**.

[0076] The container flange **20** includes a substantially flat or planar lower surface **42**. The lower surface **42** defines an underside of the container flange **20**. The support flange **28** is configured and arranged to abut against the lower surface **42** of the container flange **20**, as is shown in FIG. **3**. Put another way, the container flange is seated on the support flange. The support flange **28** is attached to the lower surface **42** of the container flange **20** by a second heat sealing material (not shown). The second heat sealing material may be a non-plastic heat sealable material, such as polylactic acid. In some arrangements, the second heat sealing material may be an aqueous heat sealing material, for example aqueous dispersions of low-density polyethylene and/or ethylene vinyl acetate. In some arrangements, the second heat sealing material may be polymeric film such as polyethylene, polypropylene, polyethylene terephthalate. In some arrangements, the second heat sealing material may be a metal foil such as aluminium foil. The second heat sealing material may be the same or different to the first heat sealing material. It will be appreciated that the support flange **28** may be coated with the second heat sealing material, or the lower surface **42** of the container flange **20** may be coated with the second heat sealing material. The support **14**, i.e. the support flange **28**, is attached to the lower surface **42** by the application of heat and pressure.

[0077] The flexible container lid **36** and the support flange **28** are attached to opposing upper and lower surfaces **34, 42** of the container flange **20** via heat sealing materials. This arrangement enables the flexible container lid **36** and the support flange **28** to be secured to the container flange **20** in a single step by applying heat and pressure to the upper and lower surfaces **34, 42** of the container flange **20**.

[0078] Although not illustrated, the container assembly of Figure include a flexible container lid, as is discussed with reference to FIGS. **1** to **4**.

[0079] Referring now to FIG. **7**, the container flange **20** of the second embodiment includes first

and second lengthwise portions **20A**, **20B**, which extend parallel to the length of the container **12**; first and second width-wise portions **20C**, **20D** which extend parallel to the width of the container **12**; and first, second, third and fourth corner portions **20E**, **20F**, **20G**, **20H** which extend between and connect the lengthwise and width-wise portions **20A**, **20B**, **20C**, **20D**, to form a continuous container flange **20**. The mouth **24** of the container **12** defined by the container flange **20** has eight sides corresponding to the eight wall panels of the container wall **18**. These include: first and second sides **62A**, **62B** which extend parallel to the length of the container **12**; third and fourth sides **62C**, **62D** which extend parallel to the width of the container **12**; a fifth side **62E** which extends between and connects the first and the third side **62A**, **62C**; a sixth side **62F** which extends between and connects the second and the fourth sides **62B**, **62D**; a seventh side **62G** which extends between and connects the first and the fourth sides **62A**, **62D**; and an eighth side **62H** which extends between and connects the second and the third sides **62B**.

[0080] It will be appreciated that the first and second sides **62A**, **62B** of the container mouth **24** are defined respectively by the first and second lengthwise portions of the container flange **20A**, **20B**; the third and fourth sides **62C**, **62D** of the container mouth **24** are defined respectively by the third and fourth width-wise portions of the container flange **20C**, **20D**; the fifth side **62E** of the container mouth **24** is defined by the first corner portion of the container flange **20E**; the sixth side **62F** of the container mouth **24** is defined by the second corner portion of the container flange **20F**; the seventh side **62G** of the container mouth **24** is defined by the third corner portion of the container flange **20G**; and the eighth side **62H** of the container mouth **24** is defined by the fourth corner portion of the container flange **20H**.

[0081] Containers **12** according to the second embodiment are designed to be supplied to food service facilities in a stacked configuration, and later de-nested and assembled with the supports **14** on site. To facilitate efficient de-nesting, the containers **12** are provided with features which interfere with corresponding features on adjacent containers **12** when the containers **12** are stacked in an alternating ABAB arrangement in which each container **20** is rotated 180 degrees relative to the adjacent containers **20** in the stack. Put another way, the containers **12** are provided with an asymmetric stacking formation. The asymmetric stacking formation may include a plurality of asymmetric recesses or depressions. In some embodiments, the container flange **20** is provided with depressions **60A-D** which interfere when the containers **12** are stacked in an ABAB arrangement so as to increase a separation between adjacent containers **12** in the stack. It will be appreciated that in alternative embodiments, two, three, five or any suitable number of depressions may be provided. Providing a series of recesses and corresponding projections help to reduce the area of interference between the containers **12** when they are stacked together, which facilitates separation of the two containers, for example by an automatic destacking device.

[0082] The first, second, third and fourth depressions **60A**, **B**, **C**, **D** are each positioned at a respective corner portion of the container flange **20E**, **20F**, **20G**, **20H**. The first, second, third and fourth depressions **60A**, **B**, **C**, **D** are contiguous with, and extend outwards from, the fifth, sixth, seventh and eighth sides **62E**, **62F**, **62G**, **62H** of the container mouth **24**, respectively. The depressions **60A**, **B**, **C**, **D** extend through the thickness of the container flange **20** such that both the upper and lower surfaces of the container flange **20** at the depressions **60A**, **B**, **C**, **D** is recessed relative to the rest of the container flange **20**.

[0083] In order that the depressions **60A**, **B**, **C**, **D** of adjacent containers **12** in the stack interfere, the positioning of the depressions **60A**, **60D** on the first and fourth container flange corner portions **20A**, **20D** differs from the positioning of the depressions **60B**, **C** on the second and third container flange corner portions **20B**, **20C**. That is, the positioning of the depressions **60A**, **D** on the container flange corner portions adjacent the third width-wise portion of the container flange **20B** differs from the positioning of the depressions **60B**, **60C** on the container flange corner portions adjacent the fourth width-wise portion of the container flange **20C**. Specifically, the first and fourth depressions **60A**, **60D** are positioned towards “outer” ends of the fifth and eighth sides, respectively,

where the term “outer” here refers to an end of the respective side which is relatively further away from a longitudinal centre line X of the container **12**. On the other hand, the second and third depressions **60C,60E** are positioned towards “inner” ends of the sixth and seventh sides **62F,62G**, respectively, where the term “inner” here refers to an end of the respective side which is relatively further away from a longitudinal centre line X of the container **12**. As a result, the depressions **60A,B,C,D** of adjacent containers **12** in the stack do not coincide when the containers **12** are stacked in the alternating ABAB configuration. Instead, the depressed lower surfaces of the depressions **60A,B,C,D** abut against the upper surface of the flange **20** of the container **12** directly below in the stack.

[0084] The container assembly **10** of the second embodiment may, if desired, be stacked in assembled form with the containers **12** inserted in the supports **14**. Accordingly, stacks of pre-assembled container assemblies **10** may be provided to a food service facility, thus removing the need for the container assemblies **10** to be assembled on site. To facilitate stacking of the pre-assembled container assemblies **10**, the base **16** of the container **12** of the second embodiment is provided with a number of base ribs **58**. The base ribs **58** act to maintain the separation between the base **16** of any given container **20** and the base **32** of the support **14** of the adjacent container assembly **10** in the stack. This improves the ease of de-nesting separate container assemblies **10**. The base ribs **58** also provide the added benefit of supporting solid food items above any juices which may accumulate in the bottom of the container during transport and storage of the packaged food items, thus helping to keep food fresh.

[0085] Although the teachings have been described above with reference to one or more preferred embodiments, it will be appreciated that various changes or modifications may be made without departing from the scope as defined in the appended claims.

Claims

1. A container assembly comprising: a plastic container comprising a base, and a container wall extending from and surrounding the base and terminating at a container flange extending around the container wall to define a mouth of the container; a support comprising a support wall arranged to surround the container wall, said support wall terminating at a support flange; and a flexible container lid attached to an upper surface of the container flange by a first heat sealing material, wherein the support flange is configured and arranged to abut against a lower surface of the container flange, and wherein the support flange is attached to the lower surface of the container flange by a second heat sealing material.
2. The container assembly according to claim 1, wherein the container flange comprises a downwardly projecting lip extending from a peripheral edge of the container flange in a direction towards the base, and wherein the lip opposes the container wall to define a channel therebetween receiving the support flange therein.
3. The container assembly according to claim 2, wherein the support flange is mechanically secured between the lip and the container wall.
4. (canceled)
5. The container assembly according to claim 1, wherein the support comprises a tab projecting from a distal edge of the support flange so as to extend beyond a distal end of the lip.
6. The container assembly according to claim 1, wherein the container wall comprises a plurality of container wall panels defining corner regions at an intersection between adjacent container wall panels and the base, and wherein each corner region is tapered or chamfered.
7. The container assembly according to claim 1, wherein the container wall comprises eight wall panels.
8. The container assembly according to claim 1, wherein the container wall comprises a plurality of container wall panels including a first pair of opposing side wall panels, a second pair of opposing

end wall panels, and four corner container wall panels, each corner container wall panel extending between a side wall panel and an adjacent end wall panel, optionally wherein container wall panels define a substantially octagonal container.

9. The container assembly according to claim 7, wherein the container flange defines a perimeter that is substantially rectangular; optionally, wherein the perimeter defines a rectangle with curved corners.

10. The container assembly according to claim 1, wherein the support wall comprises a plurality of support wall panels, and wherein the support flange comprises a support flange part extending from a distal edge of each support wall panel.

11. The container assembly according to claim 10, wherein the support flange parts are configured and arranged such that they form a continuous support flange around the support wall.

12. The container assembly according to claim 1, wherein the support wall comprises a plurality of support wall panels including a first pair of side support wall panels, a second pair of end support wall panels, and four corner support wall panels each corner support wall panels extending a support side wall panel and an adjacent support end wall panel; optionally wherein the support wall panels define a substantially octagonal support.

13. The container assembly according to claim 1, wherein the support wall comprises eight support wall panels.

14. The container assembly according to claim 12, wherein the support flange defines a perimeter that is substantially rectangular; optionally, wherein the perimeter defines a rectangle with curved corners.

15. The container assembly according to claim 1, wherein the container wall is substantially linear in a direction between the base and the container flange and tapers in a direction towards the base.

16. The container assembly according to claim 15, wherein the container wall tapers at an angle in the range 5° to 19°, in the range 10° to 18°, in the range 12° to 16°, for example approximately 14°.

17. (canceled)

18. (canceled)

19. (canceled)

20. The container assembly according to claim 1, wherein the first heat sealing material is selected from a polymeric film such as polyethylene, polypropylene, and/or polyethylene terephthalate.

21. The container assembly according to claim 1, wherein the second heat sealing material is selected from: a non-plastic heat sealable material such as polylactic acid; an aqueous heat-sealing material; polyethylene; polypropylene; and/or polyethylene terephthalate.

22. The container assembly according to claim 1, wherein the support is formed from a fibrous material such as paper or card.

23. (canceled)

24. The container assembly according to claim 1, wherein the support wall comprises a pair of support end wall panels and a pair of support side wall panels and a support corner panel between adjacent support side and end wall panels, and wherein the support flange parts extend from the support side and project beyond the respective end or side support wall panel so as to define a continuous flange around the support wall.

25. The container assembly according to claim 1, wherein the melting temperature of the second heat sealing material is less than the melting temperature of the first heat sealing material.

26. (canceled)

27. (canceled)

28. (canceled)

29. (canceled)

30. (canceled)
